Applied Project Task 2:

Implement a GitHub Actions Trigger

Unit code: COS40007

Unit Name: Artificial Intelligence Engineering

Submission date: 03 May 2026

Studio Session No.07

Group No. CL04_G04

ACKNOWLEDGMENT OF COUNTRY

Our team acknowledges the Wurundjeri People of the Kulin Nation, the Traditional Custodians of the land where Swinburne University stands. We honour their continuing connection to land, culture, and community, and pay respect to Elders past, present, and emerging.

Repository & File Structure

The project repository was structured according to the required GitHub Actions format. The correct directory hierarchy ensures that GitHub can automatically detect and execute the workflow file.

- The. GitHub/workflows/ directory contains the workflow file (train.yml), which defines the automation pipeline.
- model.py is the main script executed during the pipeline.
- requirements.txt contains all necessary dependencies.

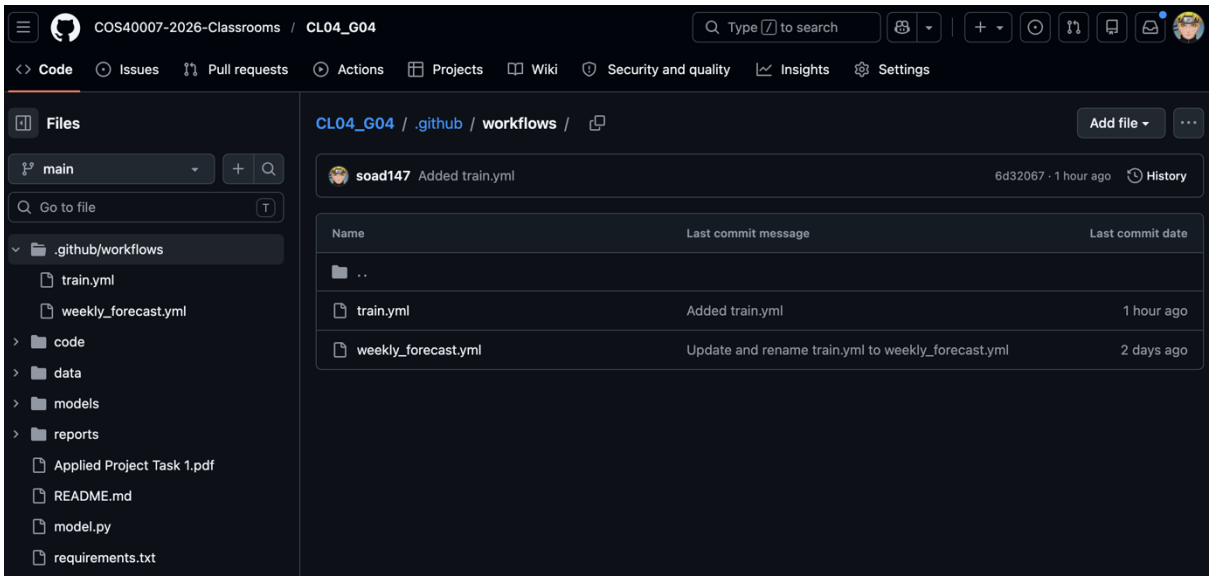

Figure 1: Repository & File structure with train.yml in. github/workflows

Requirement.txt

The requirements.txt file ensures that all necessary dependencies are installed automatically during pipeline execution. This guarantees consistency across different environments and allows the workflow to run successfully on the GitHub-hosted runner.

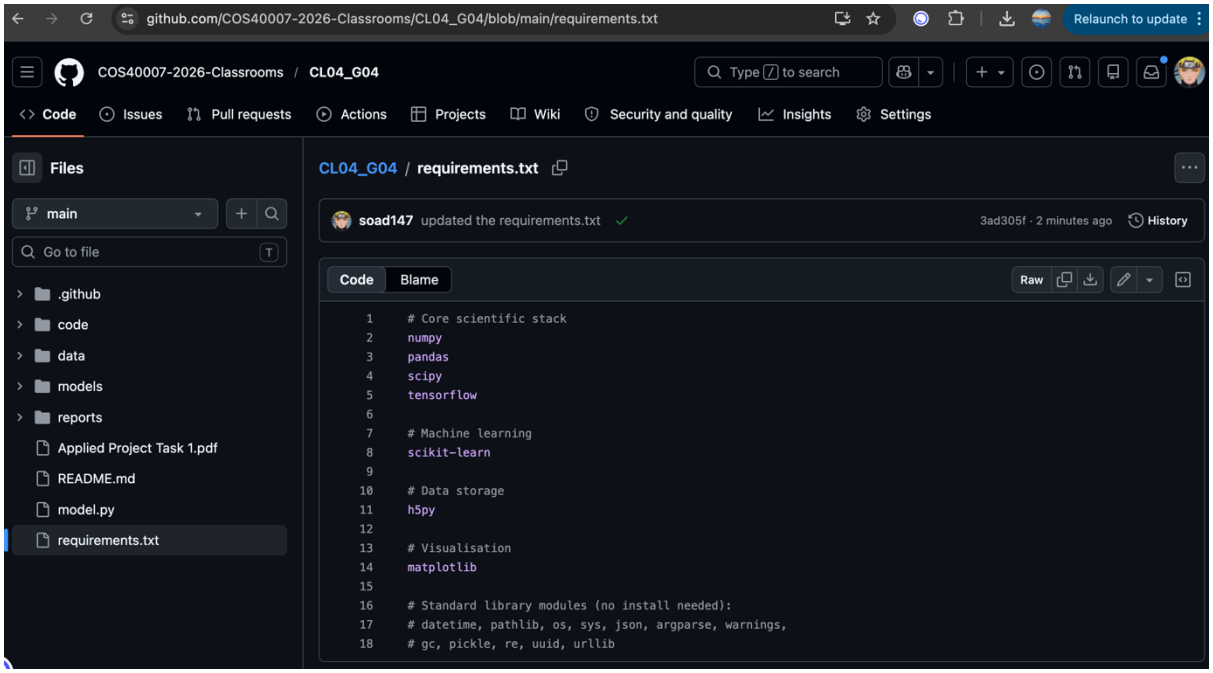

Figure 2: requirements.txt's content

Workflow triggers (on:)

The workflow is triggered automatically when changes are pushed to the main branch or when a pull request is created. Additionally, the workflow_dispatch trigger allows manual execution from the GitHub interface. This ensures flexibility in running the pipeline.

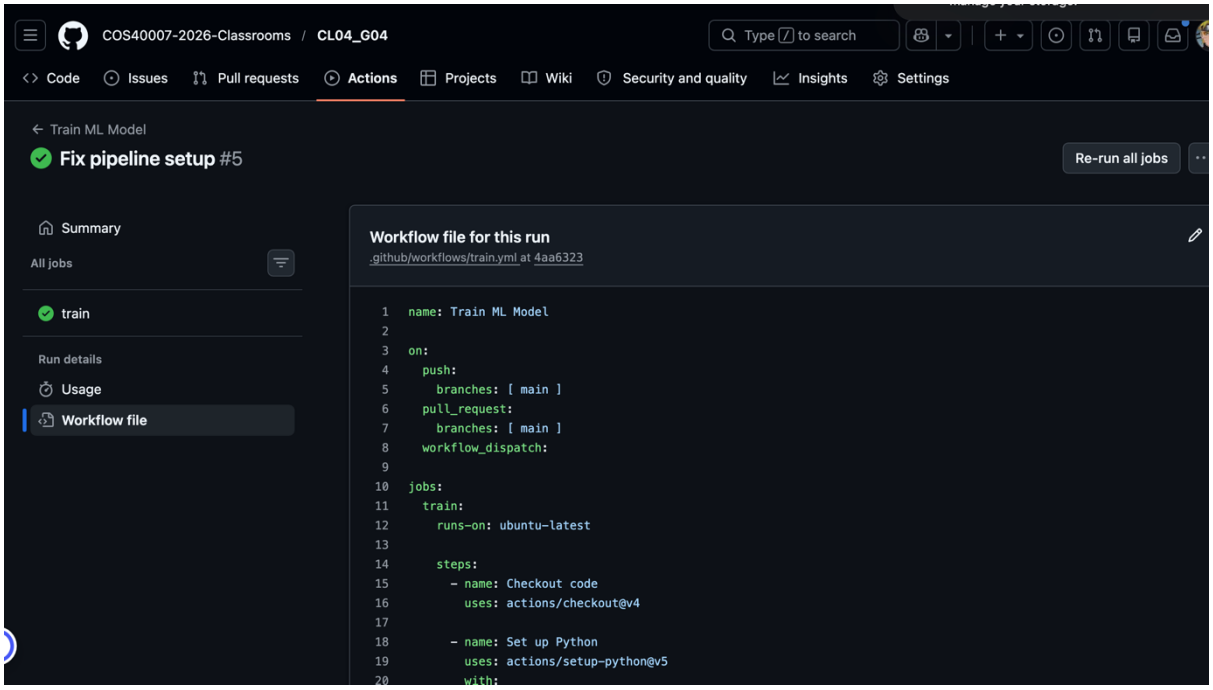

Figure 3: Workflow file

Pipeline job Definition

The workflow defines a job named train that runs on a GitHub-hosted Linux environment:

The pipeline includes all required sequential steps:

1. Checkout Code
   - Retrieves repository files for execution.
2. Set Up Python
   - Configures Python version 3.11.
3. Cache Dependencies
   - Speeds up repeated runs by caching installed packages.
4. Install Dependencies
   - Installs libraries from requirements.txt.
5. Train Model
   - Executes model.py.
6. Upload Artefacts
   - Uploads output files (model_results.png, metrics.txt).

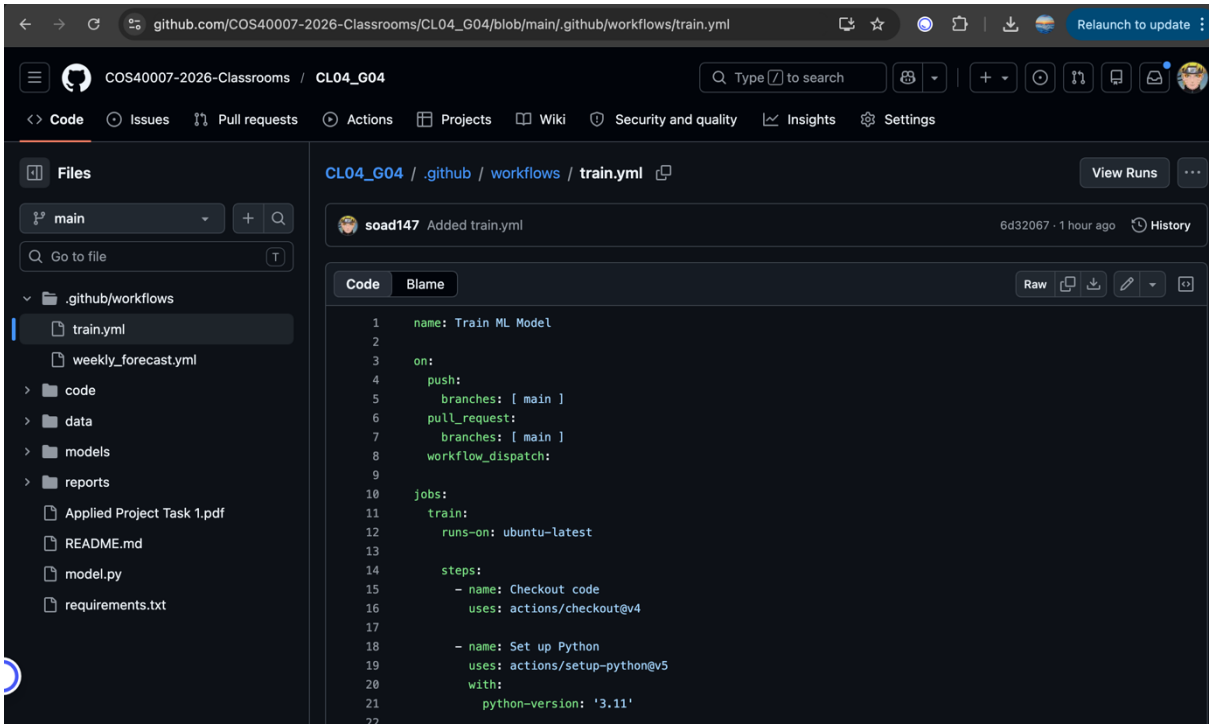

Figure 4: train.yml file

Successful Pipeline Execution

This figure shows the successful execution of the GitHub Actions workflow, where all steps completed without errors, indicated by the green status.

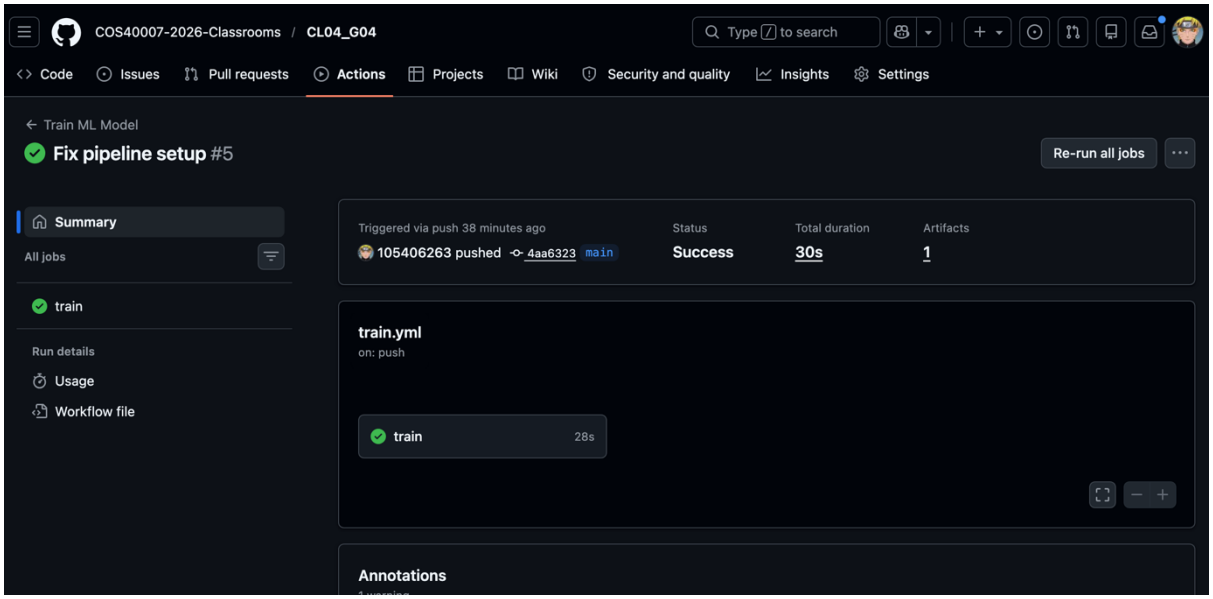

Figure 5: pipeline execution successfully

Artefact generation and upload

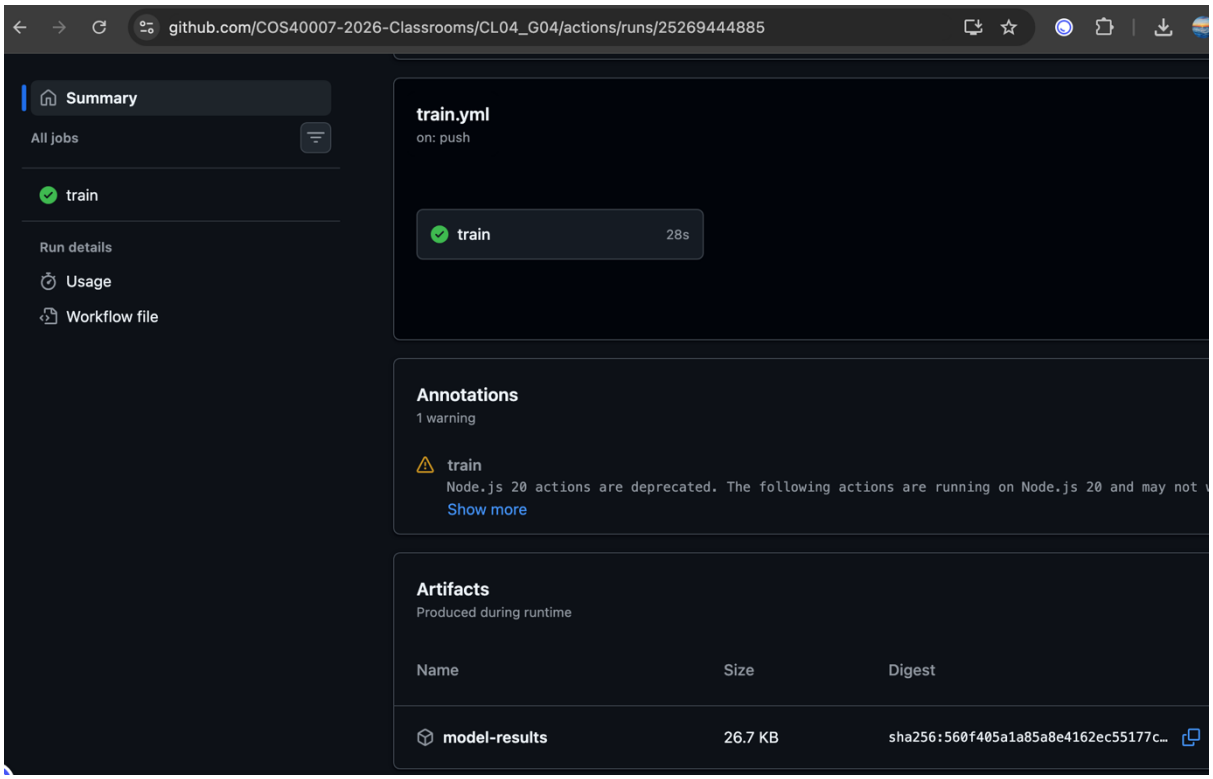

Figure 6: artefacts successfully uploaded

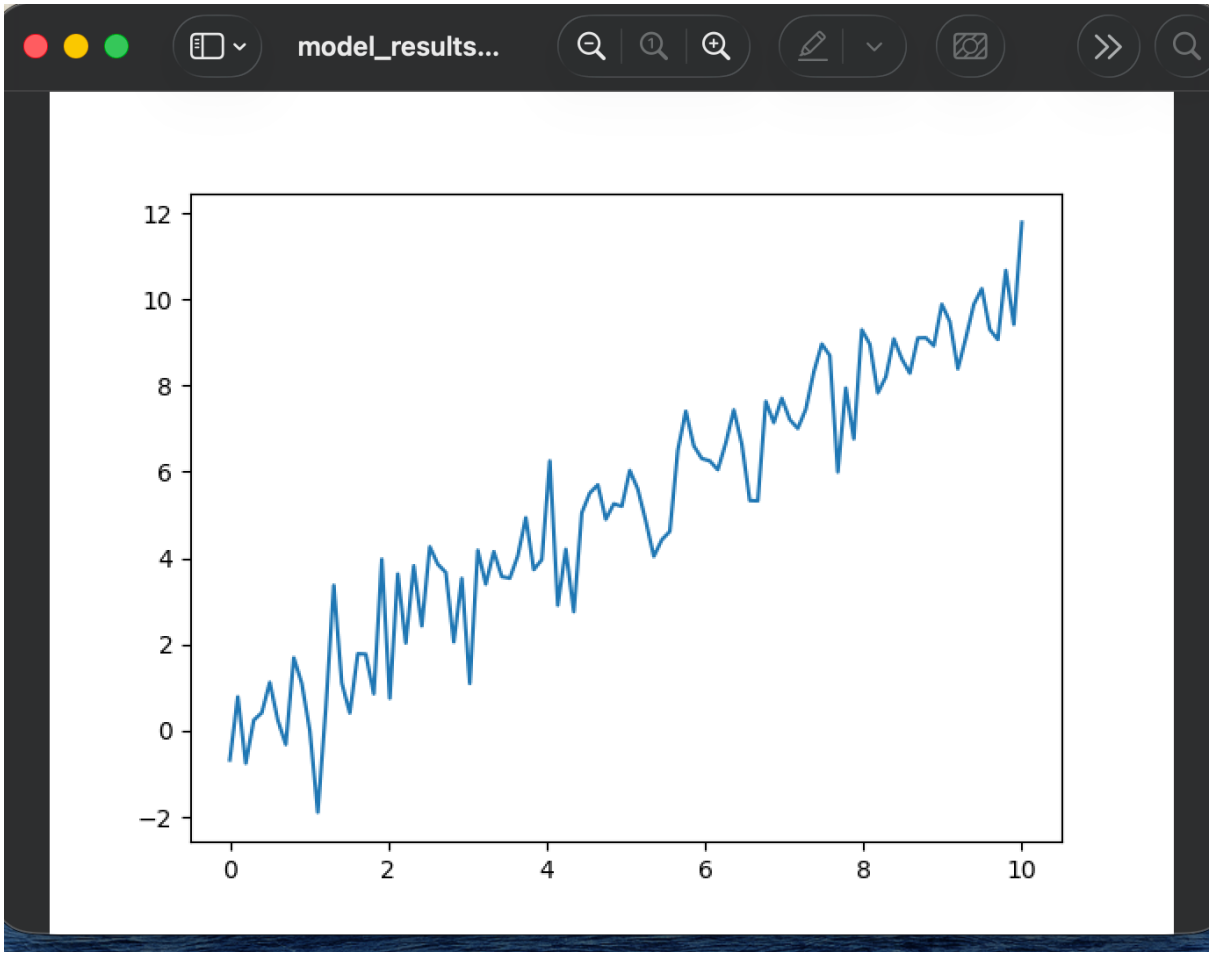

Figure 7: model_results.png

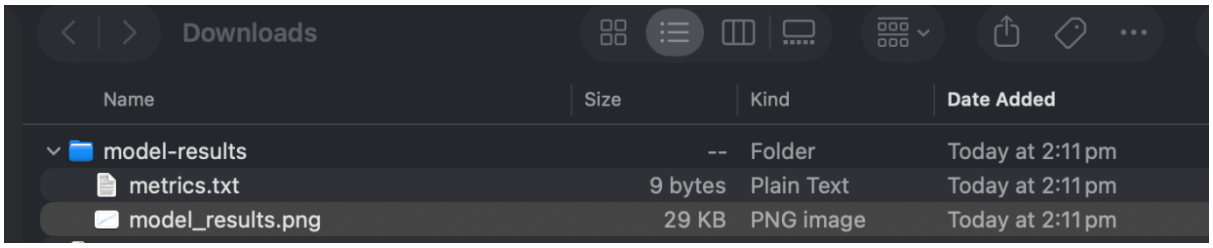

Figure 8: artefacts folder successfully downloaded

Demonstration of Functionality

The pipeline demonstrates full functionality:

- A git push triggers the workflow.
- GitHub Actions executes the pipeline automatically.
- The workflow run can be accessed via the Actions tab.
- Logs show each step being executed.
- Generated artefacts can be downloaded successfully.

Github link

The Link: https://github.com/COS40007-2026-Classrooms/CL04_G04